1. Which of the following non-linearities can produce negative outputs?
    a. sigmoid b. **tanh** c. ReLU d. softmax

2. You have to implement a feed-forward neural network having two hidden layers of dimension 512 and 256 for a 10-class classification problem. The network takes as input features of dimension 1024. How many learnable parameters will the network have?
    a. 657,920 b. 658,688 c. **658,698** d. Correct answer not present

3. What can be said about gradient-based optimization for a non-convex objective?
    a. A minima cannot be found
    b. The algorithm will not converge
    c. A minima can surely be found, but may not be a global optima
    d. An objective cannot be non-convex

4. You want to compute the gradient of a weight matrix X of dimension A * B using backprop. You receive a gradient δ from the next layer having dimension A * 1. You also compute a local gradient of γ having dimension B * 1. What would the gradient chain look like?
    a. δ * γ b. γ * δ c. **δ * γT** d. γ * δT

5. The cross-entropy loss for [.13 .29 .58]T for the reference [.1 .3 .6]T will be
    a. **0.90** b. 0.94 c. 0.39 d. Uncomputable

6. What will the gradient of [x y]T for the function 1/(1+exp(-min(x,y))) be at (x, y) = (0.2, 0.3)?
    a. [0.0 0.3]T b. **[0.25 0.0]T** c. [0.25 0]T d. [0.2 0.0]T

7. While the human eye can perceive 3 color channels, mantis shrimps can perceive 12. You want to model the mantis shrimp eye with a CNN and want to learn 30 filters of size 4 in the first layer. How many parameters will you need?
    a. **5790** b. 5760 c. 510 d. 480

8. You implemented a feed-forward neural network having one hidden dimension of size 96 that takes as input a grayscale image of 28*28 pixels. Accidentally you forgot to put a non-linearity on your hidden layer. How many parameters will you effectively lose in your implementation?
    a. **68480** b. 68384 c. 68490 d. 76330

9. You have 20 feature maps of size 6*6. You want to compute a 10-class softmax in the next layer without using any flattening. What should be the number of parameters in the CNN filters?
    a. Not possible b. **7210** c. 3610 d. 1810

10. This is a giveaway question with all answers being correct :)
On a scale of 1-4 (1 being very easy to 5 being very difficult), how difficult was today's class test? Please be honest with your answer.
    a. 1 b. 2 c. 3 d. 4